IMPLEMENTATION OF CODE CONVERTER CIRCUIT.

OBJECTIVE

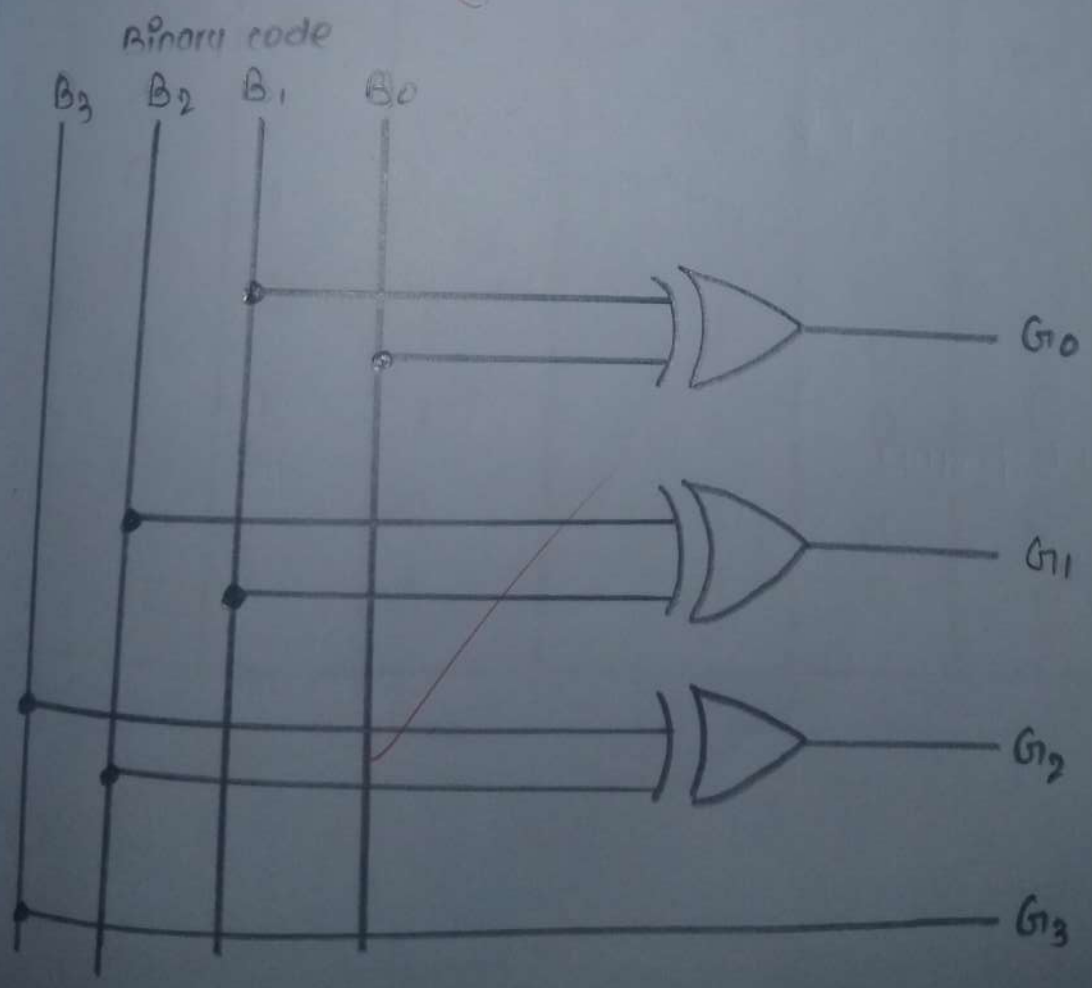
- To design and implement a digital logic circuit that converts input data from one code format to another (Binary to Gray, Gray to Binary, and BCD to Excess-3) using combinational circuit.

THEORY

Code converter

- A code converter is a circuit that makes the two systems compatible even through each uses a different binary code.

Binary to Gray code



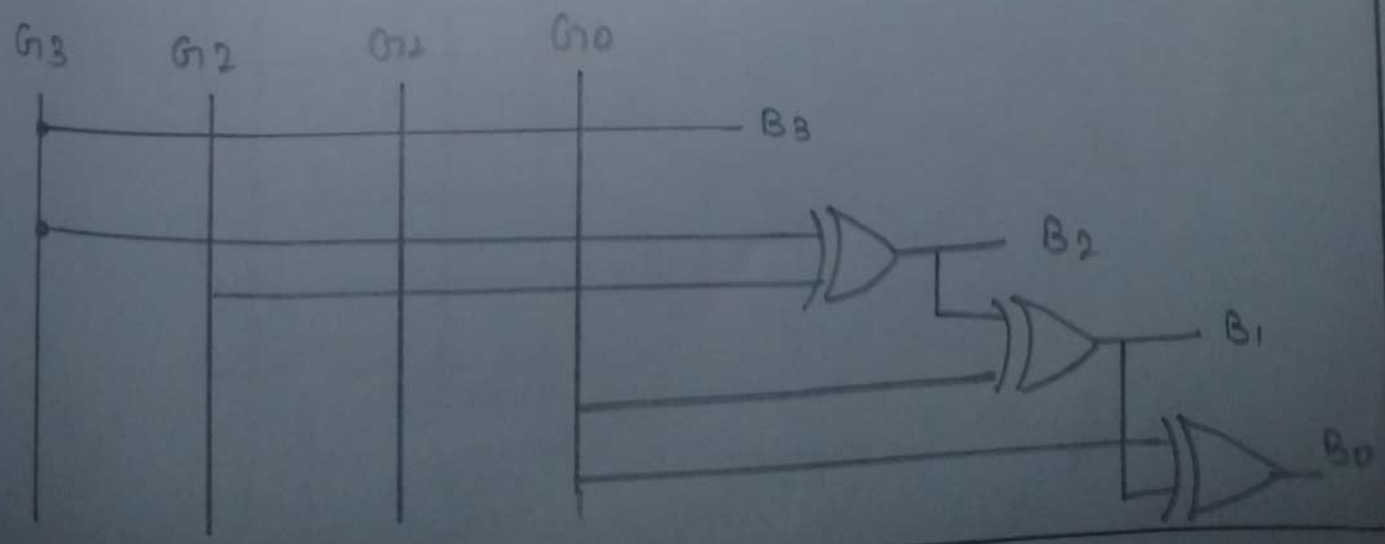
Truth table

Natural-binary code				Gray code			
B3	B2	B1	B0	G3	G2	G1	G0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

Gray-code to binary

Gray code				Natural binary code			
G3	G2	G1	G0	B3	B2	B1	B0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
1	0	0	0	1	1	1	1
1	0	0	1	1	1	1	0
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	0	1	0	1	1
1	1	1	1	1	0	1	0

Gray code

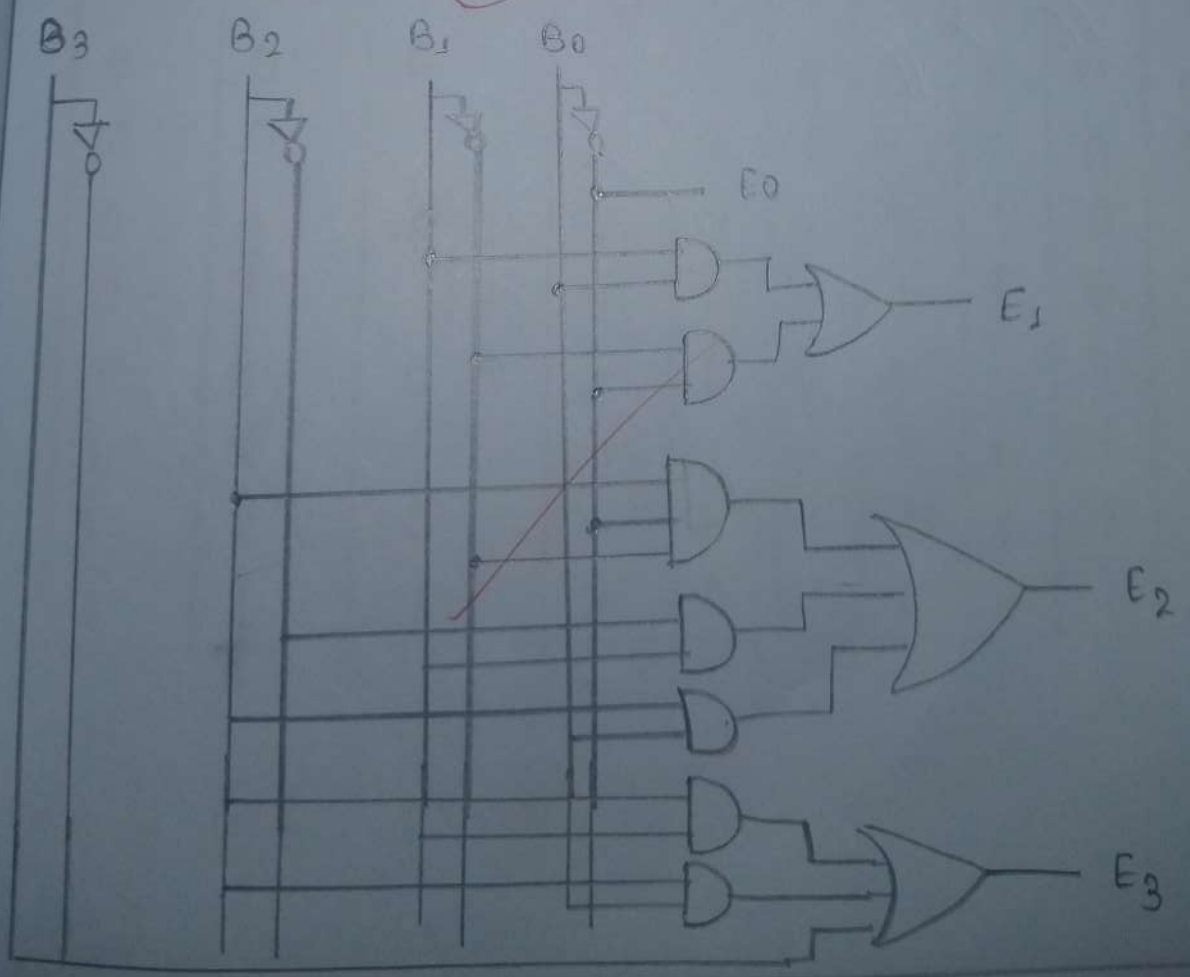


CD to Excess - 3

uth table

	BCD				Excess - 3			
ecimal	B_3	B_2	B_1	B_0	E_3	E_2	E_1	E_0
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0

Logic Diagram



BSERVATION AND DIAGRAMS

inary to Gray code

uth table

Natural-binary code				Gray code							
Inputs				Expected outputs				Observed outputs			
B3	B2	B1	B0	G3	G2	G1	G0	G3	G2	G1	G0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	0	0	0	1	0
0	0	1	1	0	0	1	1	0	0	1	1
0	1	0	0	0	1	0	0	0	1	0	0
0	1	0	1	0	1	0	1	0	1	0	1
0	1	1	0	0	1	1	0	0	1	1	0
0	1	1	1	0	1	1	1	0	1	1	1
1	0	0	0	1	0	0	0	1	0	0	0
1	0	0	1	1	0	0	1	1	0	0	1
1	0	1	0	1	0	1	0	1	0	1	0
1	0	1	1	1	0	1	1	1	0	1	1
1	1	0	0	1	1	0	0	1	1	0	0
1	1	0	1	1	1	0	1	1	1	0	1
1	1	1	0	1	1	1	0	1	1	1	0
1	1	1	1	1	1	1	1	1	1	1	1

K-map

G_3 \ B_3B_2 \ B_1B_0	00	01	11	10
00				
01				
11	1	1	1	1
10	1	1	1	1

$\therefore G_3 = B_3$

G_2 \ B_3B_2 \ B_1B_0	00	01	11	10
00				
01	1	1	1	1
11				
10	1	1	1	1

$\therefore G_2 = B_3B_2' + B_3'B_2$

$= B_3 \oplus B_2$

G_1 \ B_3B_2 \ B_1B_0	00	01	11	10
00			1	1
01	1	1		
11	1	1		
10			1	1

$\therefore G_1 = B_2B_1' + B_2'B_1$

$= B_2 \oplus B_1$

G_0 \ B_3B_2 \ B_1B_0	00	01	11	10
00		1		1
01		1		1
11		1		1
10		1		1

$\therefore G_0 = B_1'B_0 + B_1B_0'$

$= B_1 \oplus B_0$

Binary code

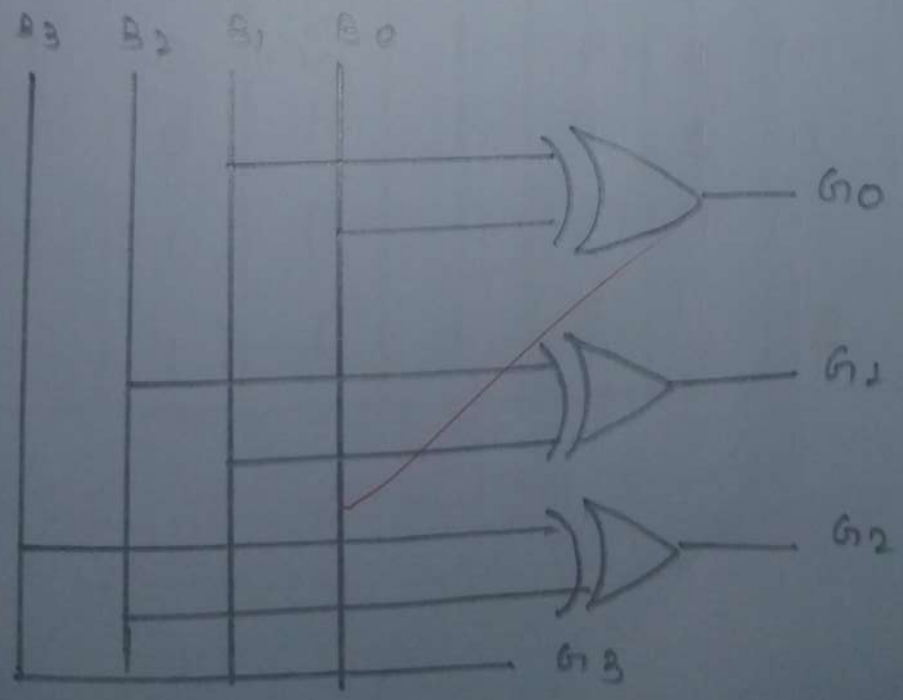


fig: logic diagram of Binary to Gray code converter

Gray code to Binary

Truth table

Input				Expected output				Observed output			
G3	G2	G1	G0	B3	B2	B1	B0	B3	B2	B1	B0
0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1	0	0	1	1
0	0	1	1	0	0	1	0	0	0	1	0
0	1	0	0	0	1	1	1	0	1	1	1
0	1	0	1	0	1	1	0	0	1	1	0
0	1	1	0	0	1	0	0	0	1	0	0
0	1	1	1	0	1	0	1	0	1	0	1
1	0	0	0	1	1	1	1	1	1	1	1
1	0	0	1	1	1	1	0	1	1	1	0
1	0	1	0	1	1	0	0	1	1	0	0
1	0	1	1	1	1	0	1	1	1	0	1
1	1	0	0	1	0	0	0	1	0	0	0
1	1	0	1	1	0	0	1	1	0	0	1
1	1	1	0	1	0	1	1	1	0	1	1
1	1	1	1	1	0	1	0	1	0	1	0

K-map

B_3	G_1G_0 = 00	01	11	10
G_3G_2 = 00				
01				
11	1	1	1	1
10	1	1	1	1

$\therefore B_3 = G_3$

B_2	G_1G_0 = 00	01	11	10
G_3G_2 = 00				
01	1	1	1	1
11				
10	1	1	1	1

$\therefore B_2 = G_3G_2' + G_3'G_2$

$= G_3 \oplus G_2$

B_1	G_1G_0 = 00	01	11	10
G_3G_2 = 00			1	1
01	1	1		
11			1	1
10	1	1		

$\therefore B_1 = G_3G_2'G_1' + G_3'G_2G_1' + G_3'G_2'G_1 + G_3G_2G_1$

$= G_3 \oplus G_2 \oplus G_1$

B_0	G_1G_0 = 00	01	11	10
G_3G_2 = 00		1		1
01	1		1	
11	1	1		1
10	1		1	

$\therefore B_0 = G_3 \oplus G_2 \oplus G_1 \oplus G_0$

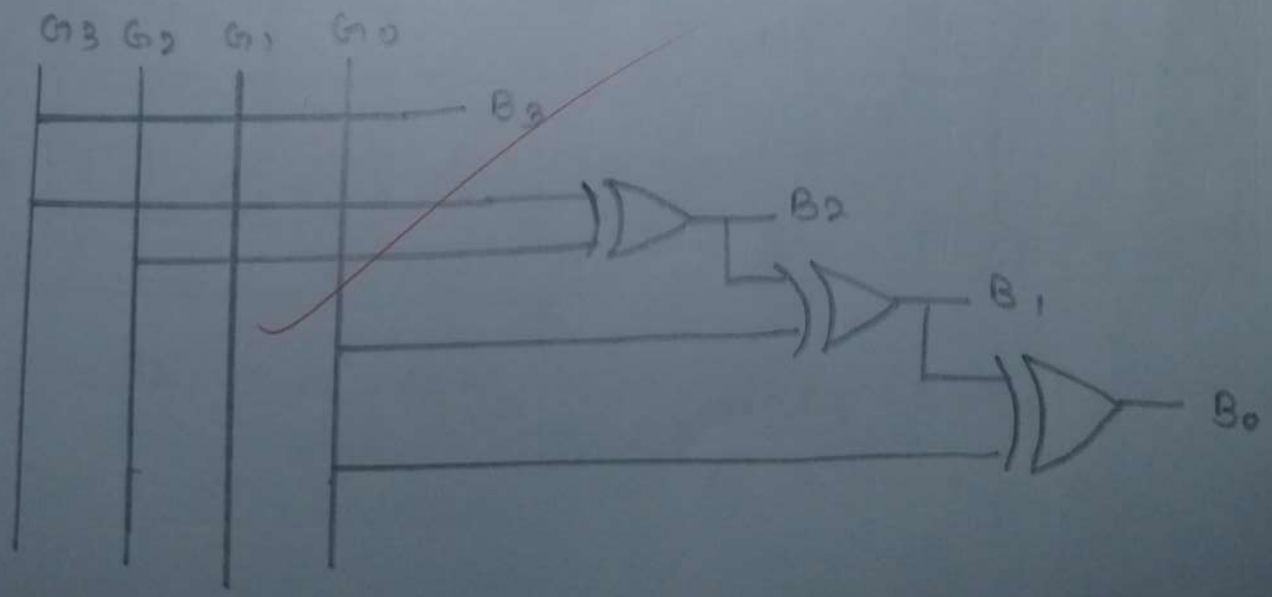


fig: logic diagram of gray code to binary converter

BCD to Excess - 3

Truth table

Decimal	BCD Inputs				Excess - 3 Expected output				Excess - 3 observed output			
	B_3	B_2	B_1	B_0	E_3	E_2	E_1	E_0	E_3	E_2	E_1	E_0
0	0	0	0	0	0	0	1	1	0	0	1	1
1	0	0	0	1	0	1	0	0	0	1	0	0
2	0	0	1	0	0	1	0	1	0	1	0	1
3	0	0	1	1	0	1	1	0	0	1	1	0
4	0	1	0	0	0	1	1	1	0	1	1	1
5	0	1	0	1	1	0	0	0	1	0	0	0
6	0	1	1	0	1	0	0	1	1	0	0	1
7	0	1	1	1	1	0	1	0	1	0	1	0
8	1	0	0	0	1	0	1	1	1	0	1	1
9	1	0	0	1	1	1	0	0	1	1	0	0

K-map

E_3

B_3B_2 \ B_1B_0	00	01	11	10
00				
01		1	1	1
11	X	X	X	X
10	1	1	X	X

$\therefore E_3 = B_3 + B_2B_1 + B_2B_0$

E_2

B_3B_2 \ B_1B_0	00	01	11	10
00		1	1	1
01	1			
11	X	X	X	X
10		1	X	X

$\therefore E_2 = B_2B_1'B_0' + B_2'B_1 + B_2'B_0$

B_3B_2 \ B_1B_0	00	01	11	10
00	1		1	
01	1		1	
11	X	X	X	X
10	1		X	X

$\therefore E_1 = B_1B_0 + B_1'B_0'$

E_0

B_3B_2 \ B_1B_0	00	01	11	10
00	1			1
01	1			1
11	X	X	X	X
10	1		X	X

$\therefore E_0 = B_0'$

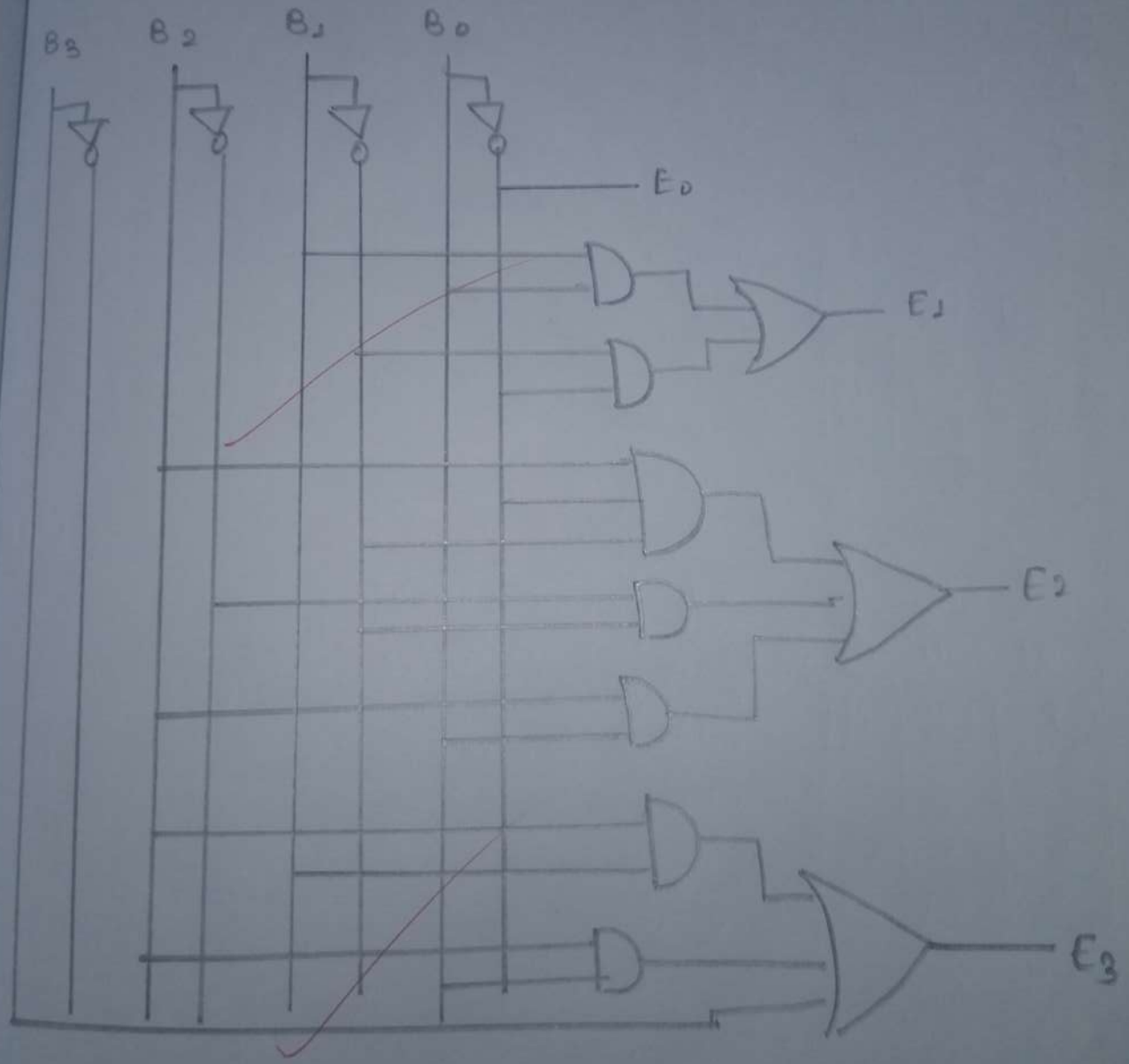


Fig: Logic diagram of binary to Excess-3 convertor

ONCLUSION

In this lab, design and implementation of a digital logic circuit that converts input data from one code format to another (Binary to Gray, Gray to binary and BCD to Excess-3) using combinational circuits was successfully implemented